CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME) Examination

January 2014

ME733 Manufacturing Planning & Control

Date: 03.01.2014, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

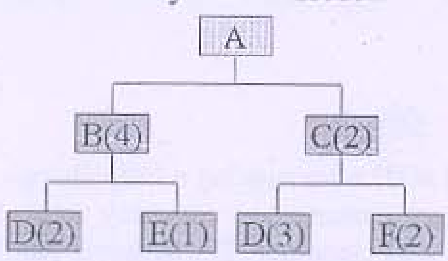
Q - 1 (a) What is planning and control in Manufacturing? [07]

Q - 2 (a) What is difference between Traditional & computer aided process planning? [08]

(b) What is Material Requirements Planning (MRP) and Manufacturing Resource Planning (MRP II)? [04]

OR

Q - 2 Given the product structure tree for "A" and the lead time and demand information below, provide a materials requirements plan that defines the number of units of each component and when they will be needed. [12]



Lead –Time days	
A	1
B	2
C	1
D	3
E	4
F	1

Demand	
Day 10	50A
Day 08	20B(SPARES)
Day 06	15D(SPARES)

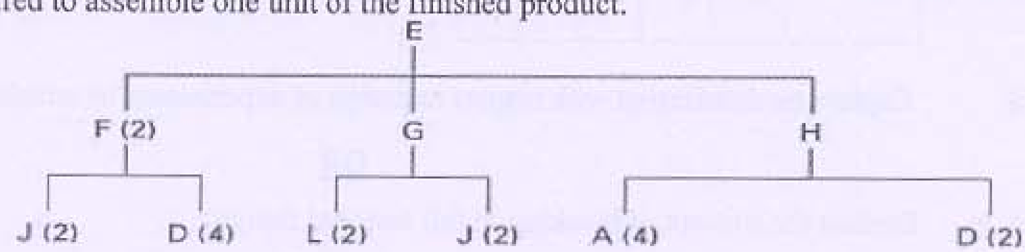
Q - 3 (a) Define Toyota Production System. What are Seven Types of Waste in it? [08]

(b) What is group Technology? Is it related with cellular manufacturing? How GT is used in Process Planning? [08]

OR

Q - 3 (a) Explain CORELAP (Computerized Relationship Layout Planning) with flow diagram. [08]

(b) Given the following diagram for a product, determine the quantity of each component required to assemble one unit of the finished product. [08]



SECTION – II

Q - 4 Write short notes on any two of the following. **[14]**
(i)kanban (ii) World Class manufacturing(iii) agile manufacturing(iv)Total Productive maintenance(v)Green production.

Q - 5 A Textile Company has three looms that weave cloth. The company is concerned that there may be significant variability in the strength of the cloth produced by the looms. Five random samples of cloths are taken from cloth produced by each looms. Each sample is tested and strength is recorded.The data are as in table. Confirm the significance from the data recorded. **[14]**

LOOMS		
1	2	3
14.0	13.9	14.1
14.1	13.8	14.2
14.2	13.9	14.1
14.0	14.0	14.0
14.1	14.0	13.9

OR

Q - 5 Company manufacturing a bearing of std 6003.after making a little change in production cycle. They receive no of complaints from customer so company wants to change the inspection method to make good relation with customer .The different part batch by the 3 different quality inspection methods are given below table. Hypothesis methods are best indicating the quality or not? ($\alpha=0.05$) **[14]**

Method	**1**	**2**	**3**	**4**	**5**
X1	2	6	8	5	4
X2	3	8	7	3	1
X3	5	4	3	2	5

Q - 6 Explain randomization with respect to design of experiments by suitable example? **[07]**

OR

Q - 6 Explain the concept of blocking in full factorial design. **[07]**

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